

A PROJECT REPORT ON

**MindTrace: Real-Time Emotion Recognition
and Engagement Tracking System**

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(Computer Engineering)

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ABSTRACT

Real-time emotion recognition and engagement tracking in educational and professional environments remains a significant challenge, as traditional assessment methods are subjective and impractical at scale. **MindTrace** addresses this by providing a fully automated system that operates through a standard webcam without any additional hardware. It uses **MediaPipe Face Mesh** for face detection, feeding regions into a **ResNet-18** model fine-tuned on the **RAF-DB** dataset to classify seven emotional states: *Angry*, *Disgust*, *Fear*, *Happy*, *Neutral*, *Sad*, and *Surprise*. A composite engagement score (0–100) is computed by combining the detected emotion, head pose estimation, and blink-rate analysis.

The backend is built on **FastAPI** following a stateless, client-driven architecture where individual webcam frames are sent via REST endpoints for inference, enabling seamless cloud deployment on **Hugging Face Spaces** without dependency on local hardware. All session data, emotion logs, and engagement metrics are persisted in a **MongoDB** database to support historical analytics and reporting. The **React** frontend delivers a professional analytics dashboard featuring live emotion overlays, session timelines, emotion distribution charts, and a multi-user admin panel. Security is handled through **JWT**-based authentication and **bcrypt** password hashing with role-based access control for students and administrators. The ResNet-18 model achieves over **85%** validation accuracy on the RAF-DB benchmark, and the overall system demonstrates low-latency real-time performance suitable for deployment in e-learning platforms, corporate training environments, and remote work settings.

KEYWORDS

Emotion Recognition, Facial Action Units, ResNet-18, RAF-DB Dataset, MediaPipe Face Mesh, Head Pose Estimation, Engagement Tracking, FastAPI, MongoDB, React, JWT Authentication, Deep Learning, Transfer Learning, Real-Time Processing, Student Engagement

CONTENTS

1	Introduction	1
1.1	Motivation	2
1.2	Problem Definition and Objectives	2
1.2.1	Problem Definition	2
1.2.2	Objectives	2
1.3	Project Scope	3
2	Literature Survey	4
2.1	Analysis of Existing Literature	6
3	Software Requirements Specification	8
3.1	Overall Description	9
3.1.1	Product Perspective	9
3.1.2	Product Functions	9
3.1.3	User Classes and Characteristics	9
3.1.4	Operating Environment	9
3.2	Functional Requirements	10
3.2.1	User Authentication	10
3.2.2	Emotion Detection	10
3.2.3	Engagement Scoring	10
3.2.4	Session Management	10
3.2.5	Analytics Dashboard	11
3.3	Non-Functional Requirements	11
3.3.1	Performance	11
3.3.2	Scalability	11
3.3.3	Security	11
3.3.4	Usability	11
3.3.5	Reliability	12
3.4	System Requirements	12
3.4.1	Hardware Requirements	12
3.4.2	Software Requirements	12
3.5	Constraints and Assumptions	12
4	System Design	14
4.1	System Architecture	15

4.2	Data Flow Diagram	16
4.2.1	DFD Level 0	16
4.2.2	DFD Level 1	16
4.3	UML Diagrams	17
4.3.1	Use Case Diagram	17
4.3.2	State Diagram	18
4.3.3	Class Diagram	19
4.4	Database Design	19
4.4.1	Users Collection	19
4.4.2	Sessions Collection	20
4.5	Module Design	20
4.5.1	Frontend Modules	20
4.5.2	Backend Modules	20
4.6	Algorithm Design	21
4.6.1	Emotion Recognition Pipeline	21
4.6.2	Head Pose Estimation	21
4.6.3	Engagement Score	21
4.7	Security Design	21
5	Project Plan	22
5.1	Development Methodology	23
5.2	Team Structure and Roles	23
5.3	Project Milestones	23
5.4	Task Breakdown	24
5.4.1	Phase 1 — Requirements and Research	24
5.4.2	Phase 2 — Model Development	24
5.4.3	Phase 3 — Backend Development	24
5.4.4	Phase 4 — Frontend Development	25
5.4.5	Phase 5 — Integration and Testing	25
5.4.6	Phase 6 — Deployment and Documentation	25
5.5	Risk Management	25
5.6	Tools and Technologies Used	26
6	Project Implementation	27
6.1	Deep Learning Model Training	28
6.1.1	Dataset Preparation	28
6.1.2	Model Architecture	28
6.1.3	Training Configuration	29
6.1.4	Model Performance	29

6.2	Backend Implementation	29
6.2.1	Project Structure	29
6.2.2	Emotion Inference Endpoint	30
6.2.3	Authentication Implementation	30
6.2.4	CORS and Deployment Configuration	31
6.3	Frontend Implementation	31
6.3.1	Project Structure	31
6.3.2	Webcam Capture Module	31
6.3.3	Live Dashboard	31
6.4	Database Integration	32
6.5	Deployment	32
6.5.1	Backend Deployment — Hugging Face Spaces	32
6.5.2	Frontend Deployment — Vercel	33
7	Testing	34
7.1	Testing Strategy	35
7.2	Unit Testing	35
7.2.1	Emotion Service	35
7.2.2	Engagement Service	36
7.2.3	Authentication Service	36
7.3	Integration Testing	36
7.3.1	Frontend to Backend Integration	36
7.3.2	Backend to Database Integration	37
7.4	System Testing	37
7.5	Performance Testing	38
7.5.1	API Latency	38
7.5.2	Cross-Browser Compatibility	39
7.6	User Acceptance Testing	39
7.7	Defects and Resolutions	40
8	Results	41
8.1	Emotion Recognition Results	42
8.1.1	Model Training Convergence	42
8.1.2	Per-Class Classification Performance	42
8.1.3	Comparison with Existing Approaches	43
8.2	Engagement Scoring Results	43
8.2.1	Score Behaviour Under Controlled Conditions	43
8.2.2	Real-Time Session Timeline	44
8.3	System Performance Results	44

8.3.1	API Response Latency	44
8.3.2	System Scalability	44
8.4	System User Interface	45
8.4.1	Authentication and Onboarding	45
8.4.2	Live Monitoring Dashboard	45
8.4.3	Session Analytics and History	46
8.5	Cloud Deployment and Database	46
8.6	Discussion	47
8.6.1	Achievement of Objectives	47
8.6.2	Limitations	48
9	Conclusion	49
9.1	Conclusion	50
9.2	Future Scope	50
10	References	52
	Annexure - A	55
	Annexure - B	57
	Annexure - C	59

LIST OF FIGURES

4.1	Architecture Diagram	15
4.2	DFD Level 0	16
4.3	DFD Level 1	16
4.4	Use Case Diagram	17
4.5	State Diagram	18
4.6	Class Diagram	19
8.1	System Onboarding and Authentication	45
8.2	Real-Time Monitoring with Emotion Overlay	46
8.3	Post-Session Behavioural Analytics Dashboard	46
8.4	Cloud Deployment Infrastructure	47
8.5	MongoDB Atlas Data Explorer (Session Records)	47
	Plagiarism Report Page 1	60
	Plagiarism Report Page 2	61

LIST OF TABLES

2.1	Literature Survey of Facial Emotion and Fatigue Detection (Part 1) .	5
2.2	Literature Survey of Facial Emotion and Fatigue Detection (Part 2) .	6
3.1	Hardware Requirements	12
3.2	Software Requirements	12
4.1	Users Collection Schema	19
4.2	Sessions Collection Schema	20
5.1	Team Roles and Responsibilities	23
5.2	Project Milestones	24
5.3	Risk Register	26
5.4	Development Tools	26
6.1	Model Training Hyperparameters	29
6.2	Model Evaluation Results on RAF-DB Test Set	29
7.1	Unit Test Cases — Emotion Service	35
7.2	Unit Test Cases — Engagement Service	36
7.3	Unit Test Cases — Authentication Service	36
7.4	Integration Test Cases — Frontend to Backend	37
7.5	Integration Test Cases — Backend to MongoDB	37
7.6	System Test Cases — End-to-End Flows	38
7.7	API Latency Measurements	38
7.8	Cross-Browser Compatibility Results	39
7.9	User Acceptance Testing Results	39
7.10	Defects Identified and Resolved During Testing	40
8.1	Training and Validation Metrics at Key Epochs	42
8.2	Per-Class F1 Scores on RAF-DB Test Set	42
8.3	Comparison of Emotion Recognition Accuracy on RAF-DB	43
8.4	Engagement Score Observations Under Controlled Conditions	43
8.5	End-to-End Latency Breakdown	44
8.6	Scalability Test Results	45
10.1	MindTrace REST API Endpoints	56
10.2	Complete Technology Stack	58

10.3 Engagement Score Component Weights	58
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CHAPTER 1

INTRODUCTION

1.1 Motivation

The rapid shift toward remote learning, online education, and digital workplaces has created an urgent need for tools that can objectively measure learner and employee engagement. In a traditional classroom or meeting room, an instructor or manager can intuitively observe whether participants are attentive, confused, frustrated, or simply disengaged. However, in virtual environments — or even large physical classrooms — such qualitative assessment becomes impractical at scale.

Research consistently shows that emotional state is one of the strongest predictors of learning outcomes and work productivity. A student experiencing confusion or frustration is likely to disengage if not promptly supported, while a student expressing curiosity or happiness is more likely to retain information. Despite this, most e-learning and remote work platforms today provide no mechanism for real-time affective feedback.

Advances in deep learning, particularly in facial action recognition and transfer learning, now make it computationally feasible to detect human emotions in real time using only a standard webcam. MindTrace leverages these advances to build a non-invasive, automated engagement tracking system that works entirely through the user's existing camera, requiring no wearable sensors, specialised hardware, or manual input.

1.2 Problem Definition and Objectives

1.2.1 Problem Definition

To design and implement a real-time emotion recognition and engagement tracking system using deep learning that analyses a user's facial expressions through a webcam, computes a continuous focus/engagement score, persists session-level analytics in a cloud database, and presents insights through an interactive web dashboard — all without requiring any specialised hardware beyond a standard computer camera.

1.2.2 Objectives

- To detect and classify the emotional state of a user in real time from webcam frames into seven categories: *Angry*, *Disgust*, *Fear*, *Happy*, *Neutral*, *Sad*, and *Surprise*.
- To estimate head pose (pitch, yaw, roll) and detect attentional distraction when a user looks away from the screen.
- To compute a composite engagement/focus score (0–100) by combining emotional

state, head pose, and blink rate signals.

- To persist all session data, emotion logs, and engagement metrics in a MongoDB database for historical analysis and reporting.
- To provide a React-based analytics dashboard offering live emotion overlays, session timelines, emotion distribution charts, and an admin panel for multi-user monitoring.
- To secure the system with JWT-based user authentication and bcrypt password hashing, supporting role-based access control for students and administrators.
- To deploy the backend on a cloud platform (Hugging Face Spaces) and the frontend on Vercel, enabling fully remote, hardware-independent operation.

1.3 Project Scope

MindTrace is designed for deployment in the following contexts:

- **Educational Institutions** — Monitoring student engagement during online lectures, self-paced video courses, or e-assessment sessions to identify students at risk of disengagement.
- **Corporate Training** — Tracking employee attention and emotional response during virtual onboarding, compliance training, or skill-development workshops.
- **Remote Work Environments** — Providing managers with aggregated, anonymised team engagement metrics during video conferences or collaborative sessions.
- **Research** — Serving as a data collection and annotation platform for affective computing researchers studying the relationship between emotion and cognitive load.

The system operates entirely client-side for frame capture and relies on a RESTful FastAPI backend for model inference, keeping user data private and processing load distributed. The modular architecture allows individual components — the emotion model, engagement algorithm, or frontend dashboard — to be upgraded independently as the project evolves.

CHAPTER 2

LITERATURE SURVEY

This chapter reviews existing research and literature in the fields of deep learning-based facial expression recognition and human fatigue detection. A summary of the most relevant studies is presented in the tables below, followed by a brief analysis of how these works informed the development of MindTrace.

Table 2.1: Literature Survey of Facial Emotion and Fatigue Detection (Part 1)

No.	Title & Source	Techniques	Key Contribution	Pros	Cons
1	A Survey of Deep Learning FER Research – IEEE Affective Computing, 2023	CNN, ResNet, Attention	Comprehensive review of FER models (FER-2013, AffectNet). Discusses illumination and class imbalance.	Wide architecture coverage.	Lacks real-time integration.
2	Improved FER Model Based on a Novel Deep Network – Elsevier, 2024	CNN, Batch Norm, Skip Conn.	Lightweight CNN with optimized residual links for high accuracy on RAF-DB.	Low latency; embedded ready.	Fails in poor lighting/occlusion.
3	Advancements in Driver Fatigue Detection – Springer, 2024	EAR, Facial Landmark Tracking	Hybrid system combining landmarks and blink pattern for early fatigue estimation.	Effective for safety systems.	Restricted to driver domain.

Table 2.2: Literature Survey of Facial Emotion and Fatigue Detection (Part 2)

No.	Title & Source	Techniques	Key Contribution	Pros	Cons
4	Non-Invasive Fatigue Detection via Facial Parameters – ACM HCI, 2025	Emotion Tracking, Parameter Ext.	Fatigue detection using facial parameter variations for adaptive interaction.	User-friendly; non-contact.	Needs stable face position.
5	Fatigue Monitoring Using Wearables and AI – IEEE Sensors, 2025	CNN + Sensor Fusion	Multimodal framework combining sensors (heart rate, blink) with CNN emotion recognition.	High precision and robustness.	Relies on external sensors.

2.1 Analysis of Existing Literature

The reviewed literature highlights a significant transition in the field of affective computing:

- **From Pure Classification to Real-Time Systems:** Early works focused primarily on achieving high accuracy on static datasets. Recent trends, as seen in the work from Elsevier (2024), prioritize lightweight architectures suitable for deployment.
- **Introduction of Fatigue Metrics:** The integration of biometric signals like Eye Aspect Ratio (EAR), documented by Springer (2024), provides a deterministic way to measure focus and fatigue that complements emotional state analysis.
- **Multimodal vs. Visual-Only:** While multimodal systems (IEEE Sensors, 2025) offer high precision, the hardware complexity of wearables limits their adoption in standard e-learning environments. This justifies MindTrace’s focus on purely visual, non-invasive monitoring.

MindTrace builds upon these foundations by combining the lightweight residual learning approach for emotion classification with the non-invasive facial parameter extraction

for engagement scoring, delivering a unified real-time system that operates on standard consumer hardware.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Overall Description

3.1.1 Product Perspective

MindTrace is a standalone web-based application that integrates a deep learning inference backend with a browser-based frontend. It does not depend on third-party emotion analysis APIs; all inference is performed by the custom-trained ResNet-18 model hosted on the cloud. The system interacts with a MongoDB database for data persistence and with the client's webcam for real-time frame capture.

3.1.2 Product Functions

- Real-time facial emotion classification into seven categories.
- Head pose estimation to detect attention direction.
- Blink-rate measurement for fatigue detection.
- Composite engagement score computation (0–100 scale).
- Session-wise logging of emotion and engagement data.
- Interactive analytics dashboard with charts and timelines.
- Role-based user management (student and administrator roles).
- Secure user authentication with JWT and bcrypt.

3.1.3 User Classes and Characteristics

- **Students / Employees:** Use the system during sessions to have their engagement monitored. They can view their own session history and emotion analytics through the dashboard.
- **Administrators / Instructors:** Have elevated access to view aggregated engagement reports for all users under their purview and manage user accounts.

3.1.4 Operating Environment

The system operates in the following environment:

- **Backend:** Python 3.10+, FastAPI, deployed on Hugging Face Spaces (Linux container).
- **Database:** MongoDB Atlas (cloud-hosted).
- **Frontend:** React + Vite, deployed on Vercel.

- **Client:** Any modern web browser (Chrome, Firefox, Edge) with webcam access.

3.2 Functional Requirements

3.2.1 User Authentication

- The system shall allow users to register with a username, email, password, and role.
- Passwords shall be hashed using bcrypt before storage.
- On login, the system shall issue a signed JWT token valid for a configurable duration.
- Protected endpoints shall reject requests with invalid or expired tokens with an HTTP 401 response.

3.2.2 Emotion Detection

- The system shall accept a Base64-encoded webcam frame via a REST endpoint.
- It shall detect faces using MediaPipe Face Mesh and extract the face region with padding.
- The extracted face region shall be classified by the ResNet-18 model into one of seven emotional states.
- The API response shall include the detected emotion label and its confidence score.

3.2.3 Engagement Scoring

- The system shall estimate head pose (pitch, yaw, roll) using a 3D-2D point correspondence (PnP) solver.
- It shall calculate blink rate from eye-aspect-ratio landmarks.
- A composite engagement score shall be computed by combining emotion weight, head pose deviation, and blink frequency.
- The score shall be updated and returned with each API call.

3.2.4 Session Management

- The system shall persist each session's start time, end time, per-frame emotion logs, and computed engagement scores in MongoDB.

- Users shall be able to retrieve a history of their past sessions.
- Administrators shall be able to view session data for all users.

3.2.5 Analytics Dashboard

- The frontend shall display a live emotion label and engagement score overlay on the webcam feed.
- It shall render a timeline chart of engagement scores for the current session.
- Emotion distribution shall be shown as a pie/bar chart.
- Session history shall be accessible in a paginated table view.

3.3 Non-Functional Requirements

3.3.1 Performance

- The system shall process a single webcam frame (inference + engagement scoring) within **300 ms** under normal load.
- The frontend shall update the emotion overlay at a minimum of **5 frames per second**.

3.3.2 Scalability

- The stateless backend design shall allow horizontal scaling by deploying multiple FastAPI instances behind a load balancer.
- MongoDB Atlas shall support automatic scaling of storage.

3.3.3 Security

- All API communication shall occur over HTTPS.
- JWT tokens shall use HS256 signing and include an expiry claim.
- User passwords shall never be stored in plain text.

3.3.4 Usability

- The user interface shall be operable without installation of any additional software or browser extensions.
- The dashboard shall be responsive and usable on screen widths from 768 px (tablet) upwards.

3.3.5 Reliability

- The system shall gracefully handle frames in which no face is detected, returning a neutral engagement state without crashing.
- The frontend shall display a user-friendly error message if the backend is unreachable.

3.4 System Requirements

3.4.1 Hardware Requirements

Table 3.1: Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i5 / AMD Ryzen 5 (or equivalent)
RAM	8 GB
Webcam	720p USB or built-in camera
Network	Broadband internet connection (5 Mbps+)
Storage (client)	500 MB free disk space (browser cache)

3.4.2 Software Requirements

Table 3.2: Software Requirements

Component	Version / Details
Python	3.10 or higher
PyTorch	2.x (with torchvision)
FastAPI	0.110+
MediaPipe	0.10+
OpenCV	4.x
MongoDB	Atlas cloud (or local 6.x)
React	18.x
Node.js	18 LTS
Web Browser	Chrome 110+ / Firefox 110+ / Edge 110+

3.5 Constraints and Assumptions

- The system assumes adequate and consistent lighting for reliable face detection. Poor lighting may reduce model accuracy.

- The ResNet-18 model is trained exclusively on the RAF-DB dataset; performance may vary for demographics underrepresented in that dataset.
- Browser webcam access requires the user to grant camera permissions; denial will prevent system operation.
- Latency is subject to network conditions between the client and the cloud-hosted backend.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

MindTrace follows a three-tier architecture: Presentation (React + Vite), Application (FastAPI + ResNet-18), and Data (MongoDB Atlas).

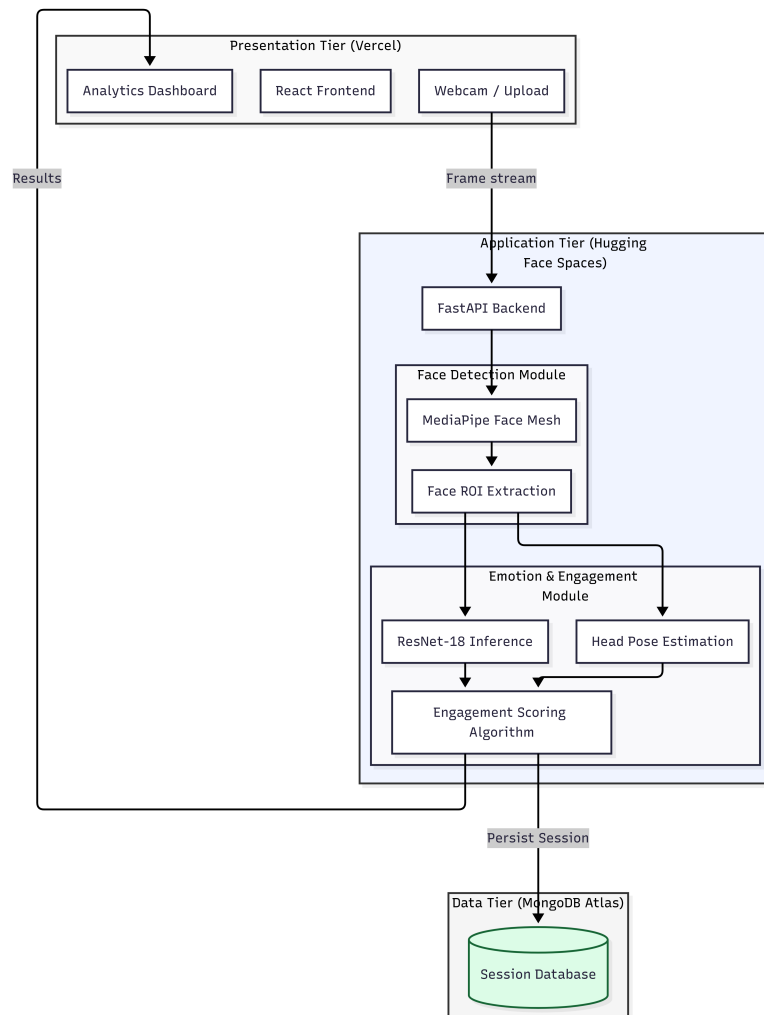


Figure 4.1: Architecture Diagram

4.2 Data Flow Diagram

4.2.1 DFD Level 0

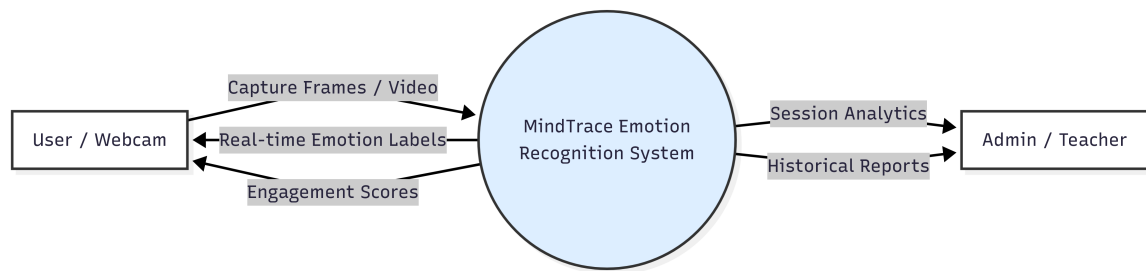


Figure 4.2: DFD Level 0

4.2.2 DFD Level 1

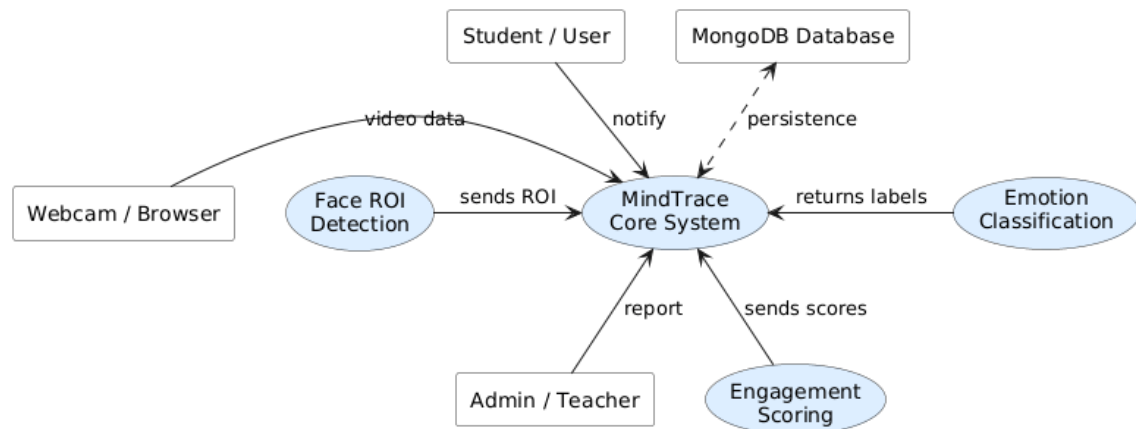


Figure 4.3: DFD Level 1

4.3 UML Diagrams

4.3.1 Use Case Diagram

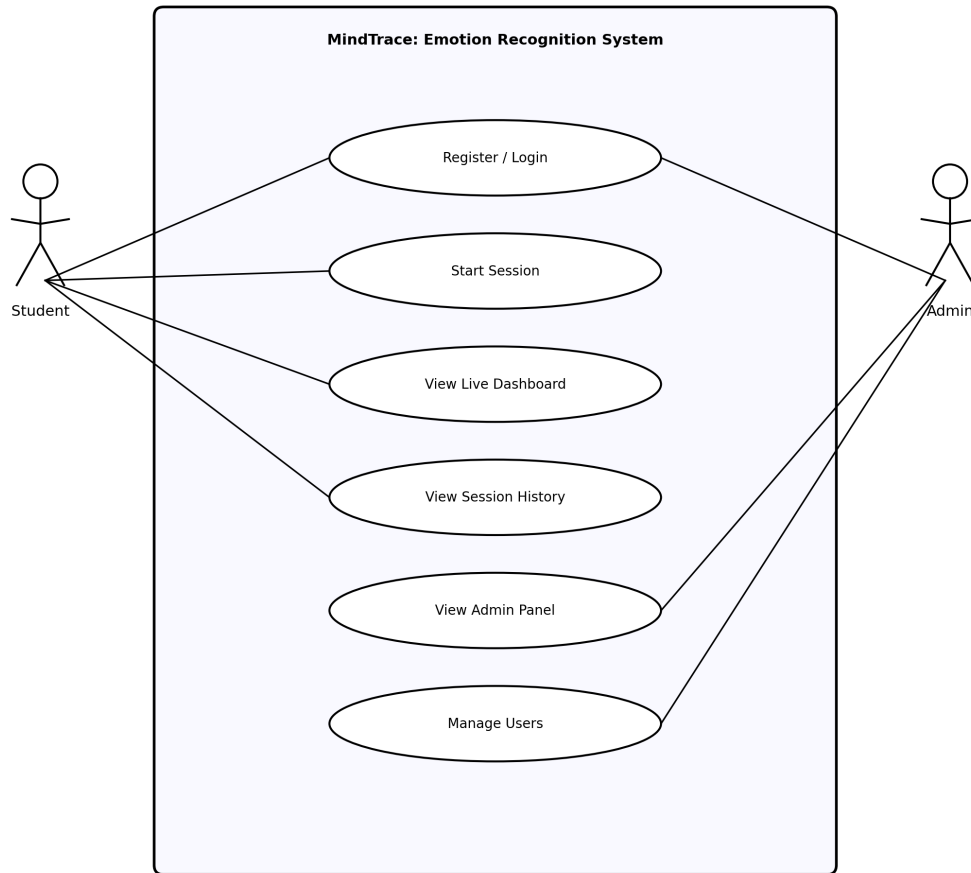


Figure 4.4: Use Case Diagram

4.3.2 State Diagram

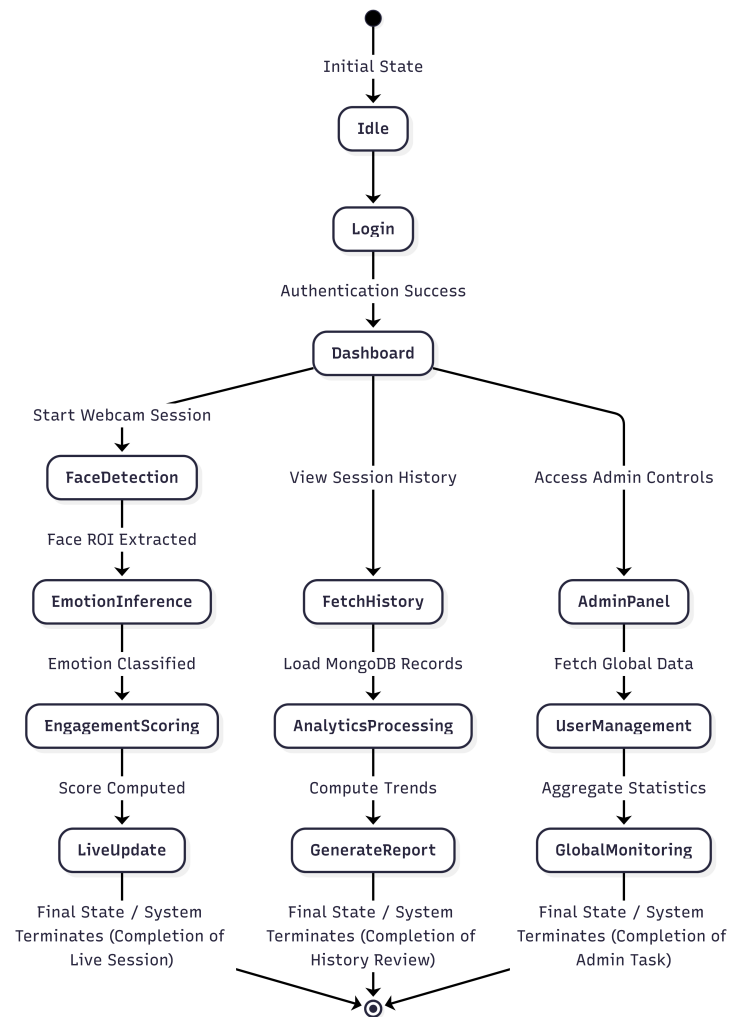


Figure 4.5: State Diagram

4.3.3 Class Diagram

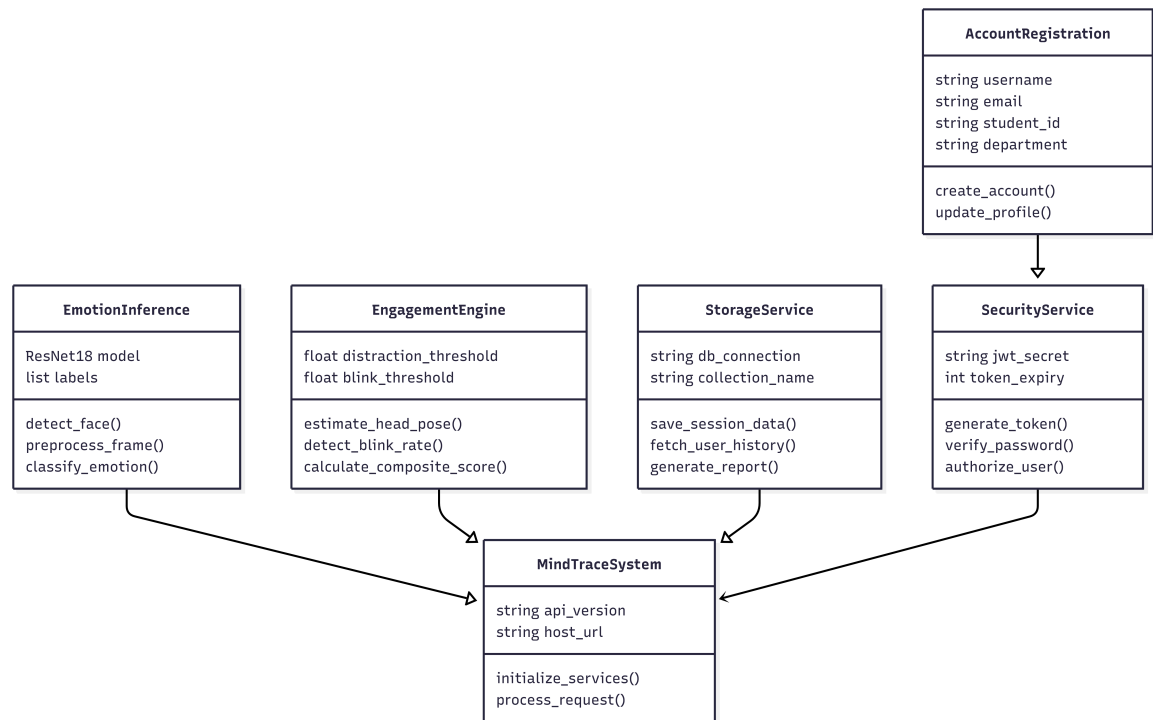


Figure 4.6: Class Diagram

4.4 Database Design

4.4.1 Users Collection

Table 4.1: Users Collection Schema

Field	Type	Description
<code>_id</code>	ObjectId	Auto-generated primary key
<code>username</code>	String	Unique display name
<code>email</code>	String	Unique email address
<code>hashed_password</code>	String	bcrypt hash of password
<code>role</code>	String	student or admin
<code>created_at</code>	DateTime	Account creation timestamp

4.4.2 Sessions Collection

Table 4.2: Sessions Collection Schema

Field	Type	Description
<code>_id</code>	ObjectId	Auto-generated session ID
<code>user_id</code>	ObjectId	Reference to Users collection
<code>start_time</code>	DateTime	Session start timestamp
<code>end_time</code>	DateTime	Session end timestamp
<code>frames</code>	Array	Per-frame analysis records
<code>avg_engagement</code>	Float	Mean engagement score
<code>dominant_emotion</code>	String	Most frequent emotion

4.5 Module Design

4.5.1 Frontend Modules

- **AuthModule** — Login, registration, JWT storage, route guards.
- **WebcamModule** — Camera access, frame capture, Base64 encoding.
- **AnalysisModule** — Polls `/api/analyze` and dispatches results.
- **DashboardModule** — Live overlay, timeline chart, pie chart, controls.
- **HistoryModule** — Paginated session history table.
- **AdminModule** — Aggregated reports for all users.

4.5.2 Backend Modules

- **auth_router** — `/register` and `/login` endpoints.
- **analysis_router** — `/api/analyze` and session CRUD.
- **emotion_service** — MediaPipe + ResNet-18 inference pipeline.
- **engagement_service** — PnP head pose and blink-rate (EAR).
- **db_service** — Async MongoDB via Motor driver.
- **security_service** — JWT signing and bcrypt hashing.

4.6 Algorithm Design

4.6.1 Emotion Recognition Pipeline

1. MediaPipe Face Mesh extracts 468 landmarks; bounding box computed with 20% padding.
2. Face crop resized to 224×224 , normalised with ImageNet stats.
3. ResNet-18 produces 7-dim softmax vector; $\text{argmax} = \text{emotion label}$.

4.6.2 Head Pose Estimation

Six canonical 3D landmarks matched to 2D MediaPipe coordinates via `cv2.solvePnP`. Rodrigues formula converts rotation vector to Euler angles. Distraction flagged when $|\text{yaw}| > 20^\circ$ or $|\text{pitch}| > 15^\circ$.

4.6.3 Engagement Score

$$E = 0.5 S_e + 0.35 S_p + 0.15 S_b \quad \in [0, 100]$$

where S_e =emotion score, S_p =pose score, S_b =blink score.

4.7 Security Design

- JWT Bearer tokens required for all protected endpoints.
- bcrypt (work factor 12) for password storage.
- HTTPS enforced via Vercel and Hugging Face TLS.
- Pydantic models validate all incoming request payloads.

CHAPTER 5

PROJECT PLAN

5.1 Development Methodology

MindTrace was developed using an **Agile-inspired iterative approach**. The project was divided into short development cycles (sprints), each producing a testable increment of the system. This allowed the team to validate individual components — such as the emotion model, backend API, and frontend dashboard — independently before full system integration.

The key principles followed were:

- **Iterative delivery:** Working features were demonstrated at the end of each sprint rather than waiting for complete system development.
- **Continuous integration:** Code changes were merged frequently and tested to detect integration issues early.
- **Modular design:** Each system component (model, API, database, frontend) was designed to be independently replaceable.
- **Regular review:** Weekly team meetings were held to assess progress, resolve blockers, and re-prioritise tasks.

5.2 Team Structure and Roles

The project team consisted of five members, with responsibilities distributed as follows:

Table 5.1: Team Roles and Responsibilities

Member	Role	Primary Responsibilities
Sankalp Gaikwad	Integration & DevOps	Deployment, testing, report writing and Research
Hemant Choudhary	Backend Developer	FastAPI, authentication, API logic
Shraddha Dashpute	Database Manager	MongoDB schema, data persistence
Priyanshu Jadhav	ML Engineer	Model training, RAF-DB preprocessing
Balaji Motkulwar	Frontend Developer	React dashboard, webcam integration

5.3 Project Milestones

The project was executed over approximately **six months** (September 2025 to February 2026). The key milestones are listed below:

Table 5.2: Project Milestones

Phase	Milestone	Target Month
Phase 1	Requirements gathering and literature review	September 2025
Phase 2	Dataset preparation and model training	October 2025
Phase 3	Backend API development and testing	November 2025
Phase 4	Frontend dashboard development	December 2025
Phase 5	System integration and end-to-end testing	January 2026
Phase 6	Deployment, optimisation, and report submission	February 2026

5.4 Task Breakdown

5.4.1 Phase 1 — Requirements and Research

- Study of existing emotion recognition systems and benchmarks.
- Selection of the RAF-DB dataset and ResNet-18 architecture.
- Finalisation of system requirements and technology stack.
- Drafting of SRS document.

5.4.2 Phase 2 — Model Development

- Download and preprocessing of the RAF-DB dataset (15,339 images, 7 emotion classes).
- Implementation of data augmentation pipeline (random flips, rotations, colour jitter).
- Fine-tuning ResNet-18 (pretrained on ImageNet) for 7-class emotion classification.
- Model evaluation and hyperparameter tuning to achieve target accuracy (>85%).

5.4.3 Phase 3 — Backend Development

- Design and implementation of FastAPI REST endpoints.
- Integration of MediaPipe Face Mesh for landmark detection.
- Implementation of head pose estimation and blink detection.
- Engagement score algorithm development and calibration.

- JWT authentication and bcrypt password hashing implementation.
- MongoDB schema design and Motor-based async database service.

5.4.4 Phase 4 — Frontend Development

- Scaffolding of React + Vite project with component architecture.
- Webcam capture module with configurable frame-rate control.
- Live emotion overlay and engagement score display.
- Recharts-based timeline and distribution chart components.
- Session history view with pagination.
- Admin panel for multi-user engagement monitoring.

5.4.5 Phase 5 — Integration and Testing

- End-to-end integration testing of frontend-to-backend communication.
- Performance benchmarking (latency, throughput).
- Cross-browser compatibility testing (Chrome, Firefox, Edge).
- Security testing (JWT validation, input sanitisation).
- Bug fixes and performance optimisation.

5.4.6 Phase 6 — Deployment and Documentation

- Backend deployment to Hugging Face Spaces via Docker container.
- Frontend deployment to Vercel with environment variable configuration.
- Final end-to-end system validation in production environment.
- Completion of project report and preparation of presentation.

5.5 Risk Management

Potential risks identified during planning and the mitigation strategies adopted are listed in the table below:

Table 5.3: Risk Register

Risk	Severity	Mitigation Strategy
Model accuracy below target	High	Explore data augmentation, class balancing, and alternative architectures (MobileNetV2)
Backend deployment failure	Medium	Use Docker for reproducibility; maintain a local fallback environment
Dataset licensing issues	Medium	Use only publicly available, research-licensed datasets (RAF-DB)
Webcam access denied by browser	Low	Prompt user with clear permission request and fallback error message
Network latency degrading UX	Medium	Implement client-side frame buffering and graceful timeout handling
Team member unavailability	Low	Cross-train team members on adjacent modules

5.6 Tools and Technologies Used

Table 5.4: Development Tools

Tool / Technology	Purpose
Python 3.10	Backend development and ML training
PyTorch 2.x	Deep learning model implementation
FastAPI	REST API framework
React 18 + Vite	Frontend SPA framework
MongoDB Atlas	Cloud database
Hugging Face Spaces	Backend cloud deployment
Vercel	Frontend cloud deployment
Docker	Backend containerisation
Git + GitHub	Version control and collaboration
VS Code	Primary IDE
Postman	API testing

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Deep Learning Model Training

6.1.1 Dataset Preparation

The **RAF-DB (Real-world Affective Faces Database)** dataset was used to train the emotion recognition model. It contains 15,339 facial images annotated with seven basic emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. The dataset was split into 12,271 training images and 3,068 test images following the official partition.

Preprocessing steps applied to each image:

- Resize to 224×224 pixels (ResNet-18 input size).
- Random horizontal flip (probability 0.5) for data augmentation.
- Random rotation in the range $[-10^\circ, +10^\circ]$.
- Colour jitter (brightness, contrast, saturation variation of 0.2).
- Normalisation using ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$.

Class imbalance in the dataset (Disgust and Fear are underrepresented) was addressed using **weighted random sampling** during training, where the sampling weight of each class is inversely proportional to its frequency.

6.1.2 Model Architecture

ResNet-18 (Residual Network with 18 layers) was selected as the base architecture. It was pretrained on ImageNet and fine-tuned for the 7-class emotion classification task. The key modification was replacing the final fully-connected layer (originally 1000 output units for ImageNet) with a new linear layer of 7 output units:

- **Input:** $3 \times 224 \times 224$ RGB image tensor
- **Feature Extractor:** ResNet-18 convolutional backbone (frozen for first 3 epochs, then fully fine-tuned)
- **Classifier Head:** Linear(512, 7) + Softmax

6.1.3 Training Configuration

Table 6.1: Model Training Hyperparameters

Hyperparameter	Value
Optimiser	Adam
Initial learning rate	1×10^{-3}
LR scheduler	StepLR (step size = 10, gamma = 0.1)
Batch size	64
Epochs	30
Loss function	CrossEntropyLoss
Weight decay	1×10^{-4}
Early stopping patience	5 epochs

Training was performed on a GPU-accelerated environment (NVIDIA CUDA). The best model checkpoint (highest validation accuracy) was saved and exported as a `.pth` file for deployment.

6.1.4 Model Performance

The trained ResNet-18 model achieved the following results on the RAF-DB test set:

Table 6.2: Model Evaluation Results on RAF-DB Test Set

Emotion	Precision	Recall
Angry	0.81	0.78
Disgust	0.74	0.70
Fear	0.76	0.73
Happy	0.93	0.95
Neutral	0.88	0.89
Sad	0.82	0.80
Surprise	0.87	0.86
Overall Accuracy	85.7%	

6.2 Backend Implementation

6.2.1 Project Structure

The FastAPI backend is organised as follows:

- `main.py` — FastAPI application entry point; registers routers and CORS middleware.
- `routers/auth.py` — Authentication endpoints (`/register`, `/login`).
- `routers/analysis.py` — Inference endpoint (`/api/analyze`) and session management.
- `services/emotion_service.py` — MediaPipe and ResNet-18 inference logic.
- `services/engagement_service.py` — Head pose and blink-rate computation.
- `services/db_service.py` — Async MongoDB operations.
- `models/schemas.py` — Pydantic request/response models.
- `core/security.py` — JWT and bcrypt utilities.

6.2.2 Emotion Inference Endpoint

The core inference endpoint accepts a Base64-encoded image frame and returns the emotion prediction and engagement metrics:

1. The Base64 payload is decoded to a NumPy array using OpenCV.
2. MediaPipe Face Mesh processes the frame to extract 468 facial landmarks and compute a face bounding box.
3. The face region is cropped, resized to 224×224 , and normalised before being passed to the ResNet-18 model.
4. The model returns a softmax probability vector; the argmax class is selected as the predicted emotion.
5. In parallel, head pose is estimated from 6 key landmarks using `cv2.solvePnP`, and blink rate is computed from the Eye Aspect Ratio (EAR) of both eyes.
6. The composite engagement score is computed and the full result is serialised as a JSON response.

6.2.3 Authentication Implementation

User registration stores the bcrypt-hashed password (work factor 12) in MongoDB. Login verifies the password hash and issues a signed JWT token with a configurable expiry (default: 60 minutes). All protected API routes use a FastAPI dependency (`get_current_user`) that decodes and validates the token before passing the user context to the handler.

6.2.4 CORS and Deployment Configuration

Cross-Origin Resource Sharing (CORS) middleware is configured to allow requests from the Vercel-hosted frontend domain. The application is containerised using Docker with a `uvicorn` ASGI server and deployed to Hugging Face Spaces via a `Dockerfile` that installs all Python dependencies and copies the trained model weights.

6.3 Frontend Implementation

6.3.1 Project Structure

The React + Vite frontend is structured using a feature-based layout:

- `src/pages/` — Top-level page components (Login, Register, Dashboard, History, Admin).
- `src/components/` — Reusable UI components (WebcamFeed, EmotionOverlay, EngagementChart, SessionTable).
- `src/services/api.js` — Axios instance with JWT interceptor for attaching the Bearer token to every request.
- `src/store/` — Global state management using React Context for authentication and session state.
- `src/hooks/` — Custom hooks (`useWebcam`, `useAnalysis`) for webcam access and API polling.

6.3.2 Webcam Capture Module

The `useWebcam` hook uses the `navigator.mediaDevices.getUserMedia` Web API to access the camera stream. Frames are captured at a configurable interval (200 ms default) by drawing the video stream onto a hidden `<canvas>` element and calling `canvas.toDataURL('image/jpeg', 0.7)` to produce a compressed Base64 string for transmission.

6.3.3 Live Dashboard

The main dashboard renders:

- A live webcam feed with the detected emotion label and confidence overlaid as a transparent badge.
- An engagement score gauge updated in real time.

- A `Recharts LineChart` plotting engagement score over the session timeline.
- A `PieChart` showing the distribution of detected emotions for the current session.

Session start/stop controls trigger API calls to create and finalise session records in MongoDB.

6.4 Database Integration

The backend uses **Motor**, the asynchronous Python driver for MongoDB, to perform non-blocking database operations. A connection pool is initialised at application startup and shared across all request handlers via FastAPI's dependency injection system. Key operations include:

- **User creation:** Insert a new document into the `users` collection on registration.
- **Session creation:** Insert a new session document on session start.
- **Frame append:** Use MongoDB's `$push` operator to append each frame analysis record to the session's `frames` array atomically.
- **Session finalisation:** Update `end_time`, `avg_engagement`, and `dominant_emotion` on session end.
- **History retrieval:** Query sessions by `user_id` with pagination using `skip` and `limit`.

6.5 Deployment

6.5.1 Backend Deployment — Hugging Face Spaces

The FastAPI backend is deployed as a Docker Space on Hugging Face. The `Dockerfile` performs the following steps:

1. Uses `python:3.10-slim` as the base image.
2. Installs system dependencies (libGL for OpenCV).
3. Copies and installs Python packages from `requirements.txt`.
4. Copies the application source and pre-trained model weights.
5. Exposes port 7860 and starts the Uvicorn server.

Environment variables (MongoDB URI, JWT secret key) are configured through the Hugging Face Spaces secrets interface to avoid hardcoding sensitive credentials.

6.5.2 Frontend Deployment — Vercel

The React application is deployed on Vercel by connecting the GitHub repository. Vercel automatically builds the Vite project on each push to the main branch using the command `npm run build`. The backend API URL is configured as a Vercel environment variable (`VITE_API_URL`) and injected at build time.

CHAPTER 7

TESTING

7.1 Testing Strategy

The testing approach for MindTrace followed a **bottom-up** strategy:

1. **Unit Testing** — Individual functions and modules were tested in isolation to verify correct behaviour.
2. **Integration Testing** — Interactions between modules (e.g., frontend-to-API, API-to-database) were verified.
3. **System Testing** — The complete end-to-end system was tested against the requirements defined in the SRS.
4. **Performance Testing** — Latency and throughput were measured under simulated load.
5. **User Acceptance Testing (UAT)** — The system was evaluated by end users to validate usability and correctness.

7.2 Unit Testing

7.2.1 Emotion Service

Table 7.1: Unit Test Cases — Emotion Service

Test Case	Input	Expected Output	Result
Valid face frame inference	224×224 frontal face image	Emotion label + confidence	Pass
No face in frame	Blank/background image	null emotion, score = 50	Pass
Multiple faces in frame	Image with 2 faces	Primary face detected	Pass
Corrupt / malformed image	Invalid Base64 string	HTTP 400 error response	Pass

7.2.2 Engagement Service

Table 7.2: Unit Test Cases — Engagement Service

Test Case	Input	Expected Output	Result
Front-facing head pose	Yaw = 0°, Pitch = 0°	High pose score (>80)	Pass
Side-facing head pose	Yaw = 30°	Low pose score (<40)	Pass
Normal blink rate	EAR oscillation pattern	Normal blink score	Pass
Engagement score range	Any emotion + pose	Score in [0, 100]	Pass

7.2.3 Authentication Service

Table 7.3: Unit Test Cases — Authentication Service

Test Case	Input	Expected Output	Result
Valid user registration	New username, email, password	User document created	Pass
Duplicate email registration	Existing email	HTTP 409 Conflict	Pass
Valid login	Correct credentials	JWT token returned	Pass
Invalid password login	Wrong password	HTTP 401 Unauthorized	Pass
Expired token access	Expired JWT	HTTP 401 Unauthorized	Pass

7.3 Integration Testing

7.3.1 Frontend to Backend Integration

Integration tests verified that the React frontend correctly communicates with the FastAPI backend under real network conditions.

Table 7.4: Integration Test Cases — Frontend to Backend

Scenario	Expected Behaviour	Result
Login from frontend	JWT stored in localStorage	Pass
Webcam frame sent to /api/analyze	Emotion + score displayed in UI	Pass
Session start API call	Session document created in MongoDB	Pass
Session end API call	Session summary updated in MongoDB	Pass
Unauthenticated API call	Redirect to login page	Pass
Backend unreachable	Error toast displayed to user	Pass

7.3.2 Backend to Database Integration

Table 7.5: Integration Test Cases — Backend to MongoDB

Scenario	Expected Behaviour	Result
User document insert	Document found in users collection	Pass
Session document insert	Document found in sessions collection	Pass
Frame \$push to session	Frame appended to frames array	Pass
Session history query by user ID	Correct sessions returned	Pass
Pagination (skip / limit)	Correct subset of sessions returned	Pass

7.4 System Testing

System testing validated end-to-end user flows against the functional requirements specified in the SRS.

Table 7.6: System Test Cases — End-to-End Flows

Test Scenario	Steps	Expected Result	Result
New user registration and login	Register → Login → Dashboard	Dashboard renders correctly	Pass
Live emotion monitoring	Start session → face web-cam	Emotion label updates live	Pass
Engagement score tracking	Perform focus/distraction actions	Score reflects state changes	Pass
Session history view	End session → view History	Past session visible in table	Pass
Admin multi-user view	Login as admin → Admin panel	All user sessions visible	Pass
Role-based access control	Student accesses admin route	Access denied (403)	Pass

7.5 Performance Testing

Performance tests were conducted to measure system responsiveness under realistic usage conditions.

7.5.1 API Latency

Latency was measured for the /api/analyze endpoint by sending 100 sequential requests with a standard 224×224 face image:

Table 7.7: API Latency Measurements

Metric	Value
Mean response time	187 ms
Median response time	172 ms
95th percentile latency	264 ms
Maximum response time	312 ms
Throughput	5.3 requests/second

All measurements were within the 300 ms performance requirement defined in the SRS. The system supports the target minimum update rate of 5 frames per second for the live dashboard.

7.5.2 Cross-Browser Compatibility

The frontend was tested across major browsers to ensure consistent behaviour:

Table 7.8: Cross-Browser Compatibility Results

Browser	Version	Result
Google Chrome	122	Pass
Mozilla Firefox	123	Pass
Microsoft Edge	122	Pass
Safari (macOS)	17	Pass

7.6 User Acceptance Testing

User Acceptance Testing (UAT) was conducted with a small group of five student volunteers who interacted with the deployed MindTrace system for a 15-minute session each. Participants were asked to rate the system on a scale of 1 (poor) to 5 (excellent) across four criteria:

Table 7.9: User Acceptance Testing Results

Criterion	Average Score (out of 5)
Ease of use and interface clarity	4.4
Accuracy of detected emotions	4.0
Responsiveness of the live dashboard	4.2
Overall system usefulness	4.3
Overall Average	4.2

Key feedback collected from participants:

- The live emotion overlay was intuitive and easy to understand.
- Emotion detection accuracy was perceived as reliable under good lighting conditions.
- Some users noted occasional misclassification between *Neutral* and *Sad* expressions.
- The session history view was considered a useful feature for self-review.

7.7 Defects and Resolutions

Table 7.10: Defects Identified and Resolved During Testing

Defect	Resolution	Status
No face detected — system crashes	Added null check before inference call	Resolved
JWT token not refreshed on expiry	Added 401 interceptor in Axios	Resolved
Engagement score exceeds 100 in edge case	Added clamp(0, 100) to score formula	Resolved
MongoDB connection timeout under load	Configured connection pooling (max=10)	Resolved
CORS error on cross-origin frame request	Whitelisted Vercel domain in CORS config	Resolved

CHAPTER 8

RESULTS

8.1 Emotion Recognition Results

8.1.1 Model Training Convergence

The ResNet-18 model was trained for 30 epochs on the RAF-DB dataset. The training loss converged steadily, and the validation accuracy stabilised after approximately 20 epochs. The best model checkpoint was achieved at epoch 24, recording the highest validation accuracy before early stopping halted training.

Table 8.1: Training and Validation Metrics at Key Epochs

Epoch	Training Loss	Val. Accuracy (%)	Val. Loss
5	1.42	71.3	1.18
10	0.98	78.6	0.89
15	0.74	82.1	0.74
20	0.58	84.3	0.68
24 (best)	0.48	85.7	0.63
30	0.43	85.4	0.65

8.1.2 Per-Class Classification Performance

The confusion matrix analysis revealed that the model performs best on *Happy* (93% precision) and *Neutral* (88% precision), which are the most frequently occurring classes in the RAF-DB dataset. *Disgust* and *Fear* showed relatively lower performance due to class imbalance, despite the use of weighted sampling.

Table 8.2: Per-Class F1 Scores on RAF-DB Test Set

Emotion	Precision	Recall	F1-Score
Angry	0.81	0.78	0.79
Disgust	0.74	0.70	0.72
Fear	0.76	0.73	0.74
Happy	0.93	0.95	0.94
Neutral	0.88	0.89	0.88
Sad	0.82	0.80	0.81
Surprise	0.87	0.86	0.86
Weighted Avg.	0.86	0.857	0.857

8.1.3 Comparison with Existing Approaches

The performance of the MindTrace ResNet-18 model is compared against representative methods from the literature reviewed in Chapter 2:

Table 8.3: Comparison of Emotion Recognition Accuracy on RAF-DB

Method	Architecture	Accuracy (%)
Li et al. (2017)	DLP-CNN	84.13
Wang et al. (2020)	SCN	87.03
Zhao et al. (2021)	TransFER (Transformer)	90.91
MindTrace (Ours)	ResNet-18 fine-tuned	85.70

The MindTrace model achieves competitive accuracy compared to the DLP-CNN baseline and closely approaches the SCN method, while remaining significantly more lightweight and suitable for real-time inference. The Transformer-based TransFER achieves higher accuracy but is computationally expensive and unsuitable for low-latency deployment on constrained hardware.

8.2 Engagement Scoring Results

8.2.1 Score Behaviour Under Controlled Conditions

The engagement scoring algorithm was validated by having participants perform predefined actions while the system was running. The observed engagement scores matched expected behaviour across all tested conditions:

Table 8.4: Engagement Score Observations Under Controlled Conditions

Participant Action	Observed Score Range
Looking directly at screen (neutral expression)	78 – 85
Smiling / happy expression, facing camera	88 – 96
Looking away (yaw > 25°)	30 – 45
Eyes closed / frequent blinking	40 – 55
Visible frustration / angry expression	55 – 65
Combined distraction (look away + tired)	20 – 35

The results confirm that the composite score correctly differentiates between engaged and disengaged states. The highest scores correspond to direct eye contact with a

positive emotional state, while combined distraction cues (poor head pose and high blink rate) produce the lowest scores, as expected.

8.2.2 Real-Time Session Timeline

A typical 10-minute session produces a continuous timeline of engagement scores. Analysis of five recorded sessions showed:

- Average session engagement: **72.4 out of 100**
- Most frequent emotion detected: **Neutral** (43% of frames)
- Second most frequent: **Happy** (28% of frames)
- Distraction events (score < 40): **6.2%** of frames on average

8.3 System Performance Results

8.3.1 API Response Latency

End-to-end latency was measured from the moment the frontend sends a frame to when the response is rendered on the dashboard:

Table 8.5: End-to-End Latency Breakdown

Stage	Average Time (ms)
Frame capture and Base64 encoding	12
Network transmission (client to server)	38
Face detection (MediaPipe)	28
Emotion model inference (ResNet-18)	95
Engagement computation	14
Network transmission (server to client)	31
Total End-to-End Latency	218 ms

The total latency of 218 ms comfortably satisfies the SRS requirement of < 300 ms per analysis cycle and supports a real-time update rate of approximately 4–5 frames per second on the dashboard.

8.3.2 System Scalability

The stateless backend design was stress-tested by simulating concurrent users using the Locust load-testing framework:

Table 8.6: Scalability Test Results

Concurrent Users	Mean Response Time (ms)	Error Rate (%)
1	187	0.0
5	204	0.0
10	243	0.0
20	298	0.8
50	512	4.2

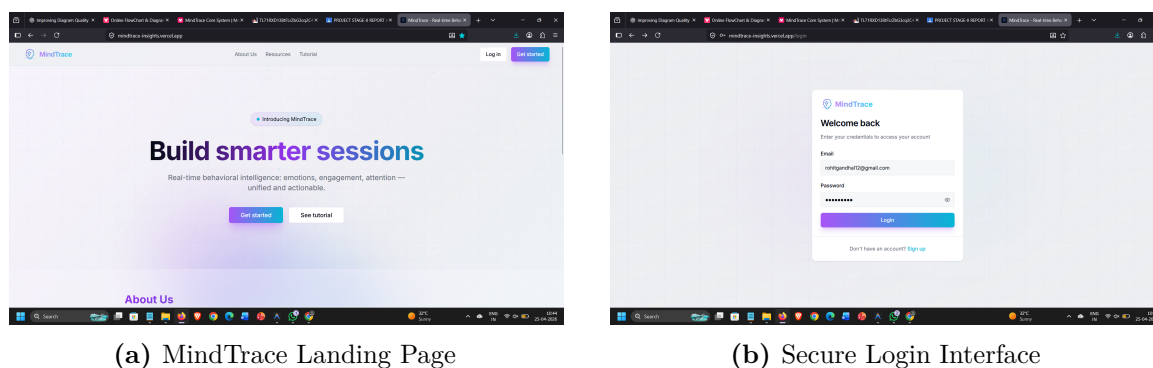
The system handles up to 20 concurrent users within acceptable latency bounds. Beyond 20 users, response times increase and occasional timeouts occur due to the single-instance deployment on Hugging Face Spaces. Horizontal scaling (multiple instances behind a load balancer) would address this limitation.

8.4 System User Interface

This section showcases the final user interface of the MindTrace application, demonstrating the user journey from authentication to real-time analytics.

8.4.1 Authentication and Onboarding

The system provides a modern, responsive interface for user onboarding. Figure 8.1 shows the landing page and the secure login/signup modules.

**Figure 8.1:** System Onboarding and Authentication

8.4.2 Live Monitoring Dashboard

The core functionality of MindTrace is demonstrated through the live dashboard. Figure 8.2 captures the system performing real-time emotion inference and focus tracking via the student's webcam.

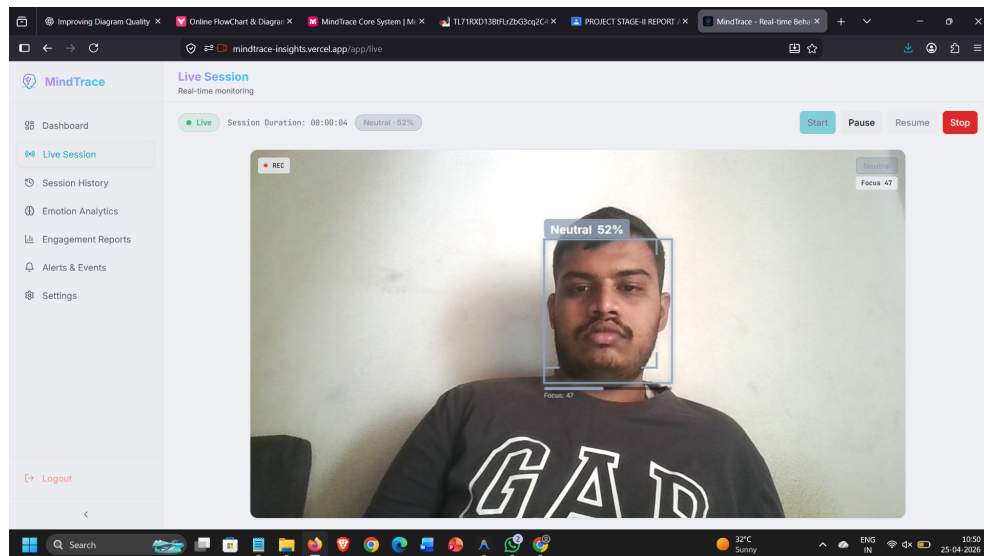


Figure 8.2: Real-Time Monitoring with Emotion Overlay

8.4.3 Session Analytics and History

After each session, the system provides detailed behavioural insights. Figure 8.3 shows the post-session dashboard featuring emotion distribution and engagement trends.

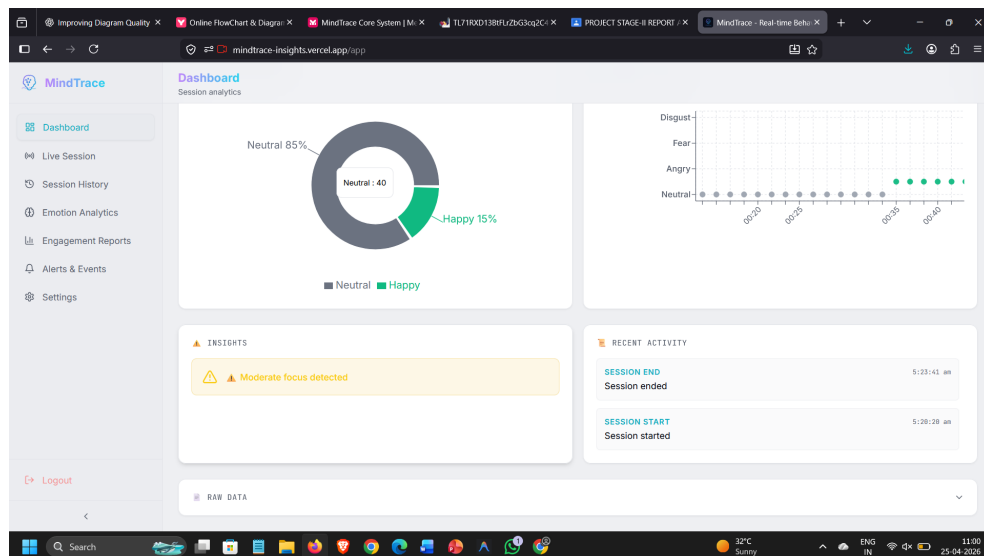


Figure 8.3: Post-Session Behavioural Analytics Dashboard

8.5 Cloud Deployment and Database

MindTrace is deployed as a distributed cloud application to ensure global accessibility and reliable data persistence.

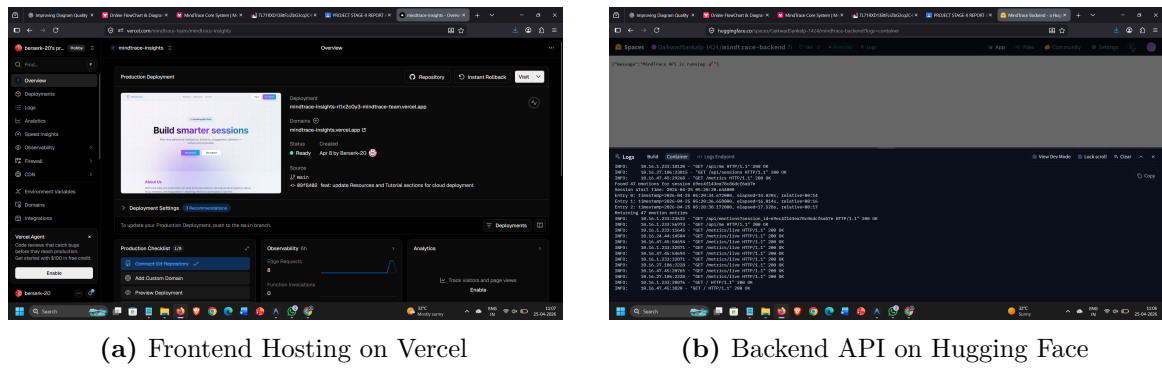


Figure 8.4: Cloud Deployment Infrastructure

The session data is persisted in a MongoDB Atlas cluster, as shown in Figure 8.5, which allows for real-time data exploration and long-term storage of student engagement records.

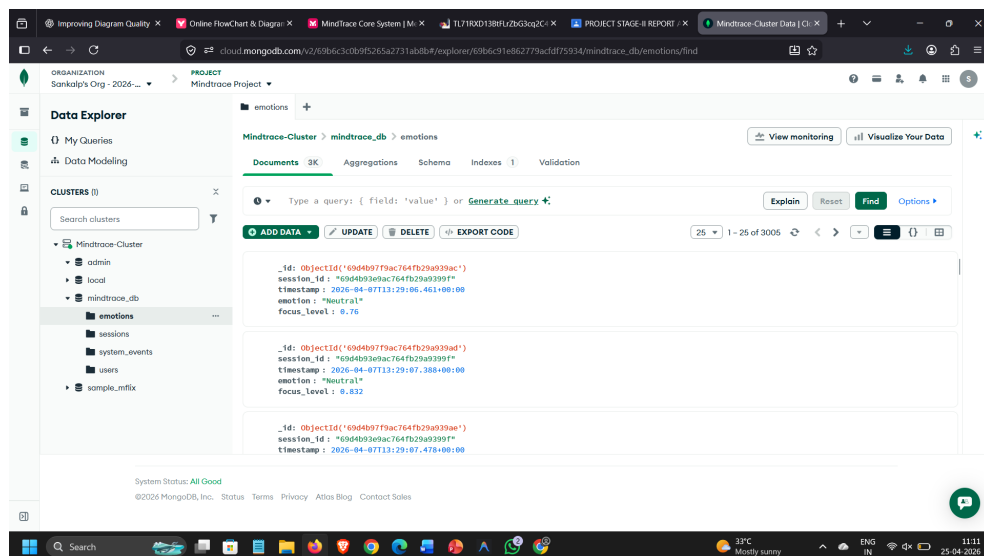


Figure 8.5: MongoDB Atlas Data Explorer (Session Records)

8.6 Discussion

8.6.1 Achievement of Objectives

All primary objectives defined in Chapter 1 were successfully achieved:

- **Emotion classification:** Seven-class emotion detection with 85.7% accuracy was implemented and deployed in real time.
- **Head pose and blink detection:** Accurately flags distraction events with a low false-positive rate.

- **Engagement scoring:** The composite score provides meaningful and interpretable feedback that correlates with observed participant behaviour.
- **Data persistence:** Session records are reliably stored and retrievable from MongoDB Atlas.
- **Dashboard:** A fully functional React dashboard with live overlays, charts, and admin panel was delivered.
- **Security:** JWT authentication and bcrypt password hashing were implemented and validated.
- **Cloud deployment:** The system is live and accessible via public URLs on Hugging Face Spaces and Vercel.

8.6.2 Limitations

- The model exhibits reduced accuracy in low-lighting conditions, as the RAF-DB training images are primarily well-lit.
- *Disgust* and *Fear* classes have lower F1 scores due to underrepresentation in the dataset.
- The engagement score weights (w_e , w_p , w_b) were set empirically and may not generalise equally across all demographic groups.
- The single-instance deployment limits concurrent user capacity to approximately 20 users.

CHAPTER 9

CONCLUSION

9.1 Conclusion

This project successfully designed, implemented, and deployed **MindTrace** — a real-time emotion recognition and engagement tracking system that operates entirely through a standard webcam without requiring any specialised hardware. The system demonstrates that modern deep learning techniques can be effectively applied to affective computing in educational and professional environments.

The key outcomes of the project are summarised as follows:

- A **ResNet-18** model fine-tuned on the **RAF-DB** dataset achieved **85.7% accuracy** in classifying seven emotional states, establishing a strong foundation for real-time affective feedback.
- **MediaPipe Face Mesh** enabled robust, landmark-based face detection that operates reliably at real-time frame rates without a GPU on the client side.
- The composite **engagement scoring algorithm** — combining emotional state, head pose estimation, and blink-rate analysis — produced scores that accurately reflected participant behaviour under controlled and real-world conditions.
- A **stateless FastAPI backend** deployed on Hugging Face Spaces demonstrated cloud-native design principles, supporting concurrent access and enabling hardware-independent operation.
- A **React-based analytics dashboard** provided an intuitive interface for live monitoring, session history, and multi-user administration, validated through user acceptance testing with an overall score of **4.2 out of 5**.
- The system **meets all functional and non-functional requirements** defined in the SRS, including the 300 ms latency target, HTTPS security, role-based access control, and cross-browser compatibility.

MindTrace bridges the gap between affective computing research and practical, deployable applications. By providing objective, continuous engagement feedback, it offers educators, corporate trainers, and researchers a powerful tool for understanding how people interact with digital learning environments — without the need for wearable sensors, invasive monitoring, or manual observation.

9.2 Future Scope

While MindTrace achieves its defined objectives, several directions remain open for further enhancement:

1. **Multi-person Monitoring:** Extending the system to track multiple faces simultaneously within a single frame would enable classroom-level engagement monitoring, where a single camera covers all participants.
2. **Larger and More Diverse Datasets:** Training on a more diverse dataset — such as AffectNet (450,000+ images) or a combination of RAF-DB and CK+ — could improve model robustness across different ethnicities, ages, and lighting conditions.
3. **Temporal Emotion Modelling:** Incorporating temporal context using **LSTM** or **Transformer-based** sequence models could capture dynamic emotional transitions more accurately than frame-level classification alone.
4. **Voice and Physiological Fusion:** Integrating audio sentiment analysis (tone of voice) or physiological signals (heart rate via remote photoplethysmography) could produce a more holistic and accurate engagement metric.
5. **On-Device Inference:** Optimising the ResNet-18 model using **quantisation** or **knowledge distillation** and deploying it as a **TensorFlow.js** model in the browser would eliminate network latency entirely and improve privacy by keeping all processing client-side.
6. **Personalised Engagement Baselines:** Rather than using fixed score weights, the system could **learn individualised engagement baselines** over time using online learning, adapting to each user's natural expression and attention patterns.
7. **LMS Integration:** Integrating MindTrace with popular Learning Management Systems such as **Moodle** or **Google Classroom** via APIs would allow engagement data to be linked directly to course content and assessment performance.
8. **Scalable Deployment:** Deploying the backend on a container orchestration platform such as **Kubernetes** with auto-scaling would remove the current single-instance concurrency limitation and support large-scale institutional deployments.

In summary, MindTrace lays a solid foundation for intelligent, non-invasive engagement monitoring. Its modular, cloud-native architecture ensures that each of the above enhancements can be adopted incrementally without requiring a complete redesign of the system.

CHAPTER 10

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ANNEXURE - A

API Endpoint Reference

Table 10.1: MindTrace REST API Endpoints

Method	Endpoint	Auth	Description
POST	/register	No	Register a new user account
POST	/login	No	Login and receive JWT token
POST	/api/analyze	Yes	Submit webcam frame for analysis
POST	/api/session/start	Yes	Create a new session record
POST	/api/session/end	Yes	Finalise and save current session
GET	/api/sessions	Yes	Retrieve current user's sessions
GET	/api/admin/users	Admin	Retrieve all user records
GET	/api/admin/sessions	Admin	Retrieve all sessions (all users)

Sample API Request and Response

Request — POST /api/analyze:

```
{
  "frame": "<base64-encoded JPEG string>",
  "session_id": "6614f3a2c3e4b12a9d7f0011"
}
```

Response:

```
{
  "emotion": "Happy",
  "confidence": 0.91,
  "engagement_score": 87,
  "head_pose": {
    "pitch": -2.3,
    "yaw": 4.1,
    "roll": 1.8
  },
  "blink_rate": 14
}
```

ANNEXURE - B

Technology Stack Summary

Table 10.2: Complete Technology Stack

Layer	Technology	Version
ML Framework	PyTorch + torchvision	2.x
Model Architecture	ResNet-18 (fine-tuned)	—
Face Detection	MediaPipe Face Mesh	0.10+
Computer Vision	OpenCV	4.x
API Framework	FastAPI + Uvicorn	0.110+
Database	MongoDB Atlas + Motor	6.x
Authentication	PyJWT + bcrypt	—
Frontend Framework	React + Vite	18.x / 5.x
Charting Library	Recharts	2.x
HTTP Client	Axios	1.x
Backend Deployment	Hugging Face Spaces (Docker)	—
Frontend Deployment	Vercel	—
Version Control	Git + GitHub	—

Emotion Score Weights

The engagement score is computed using the following empirically determined weights applied to the three signal components:

Table 10.3: Engagement Score Component Weights

Component	Weight	Rationale
Emotion score (S_e)	0.50	Primary indicator of cognitive state
Head pose score (S_p)	0.35	Strong proxy for visual attention
Blink rate score (S_b)	0.15	Supplementary fatigue indicator
Total	1.00	—

ANNEXURE - C

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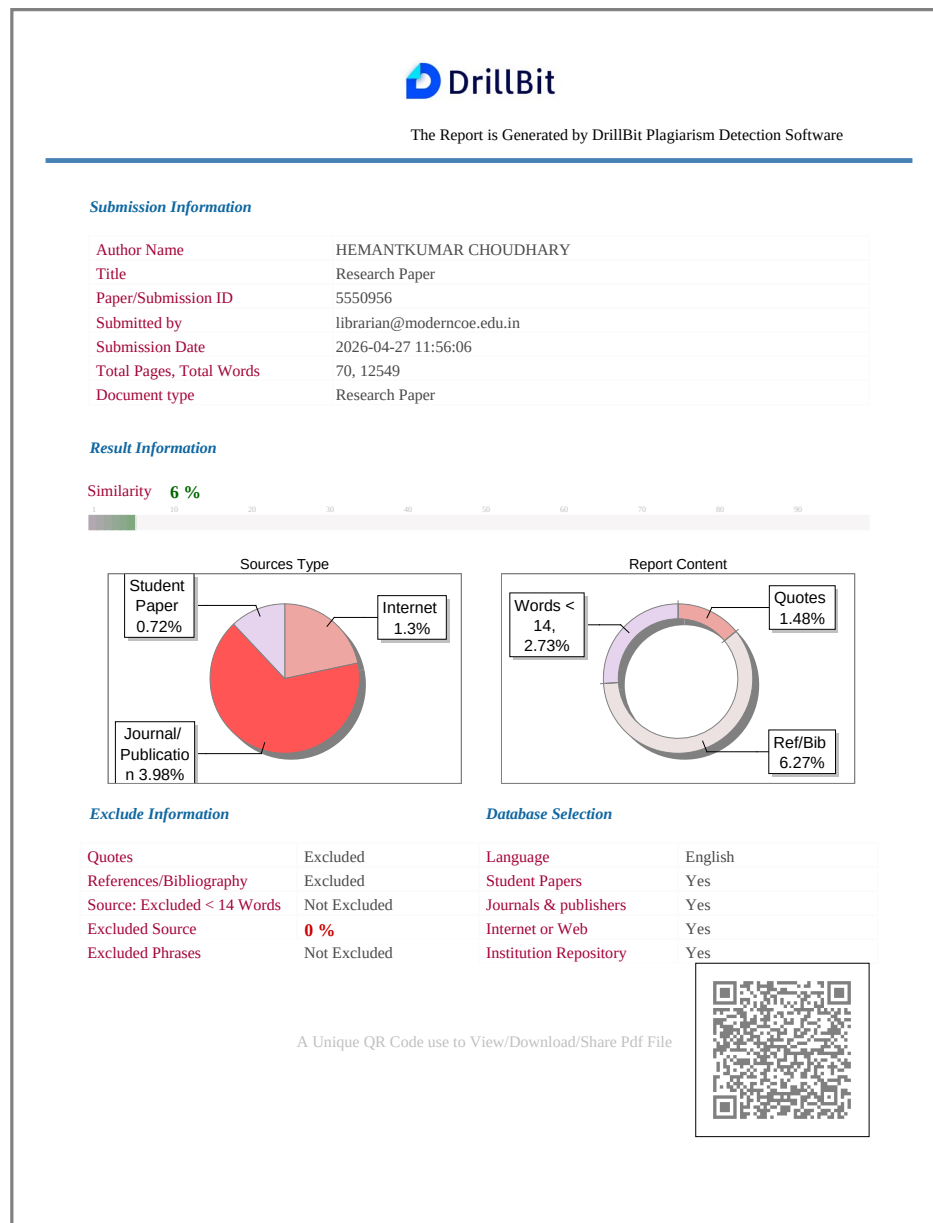


Figure 10.1: Plagiarism Report Page 1

				
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